THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VIII — SCIENCE · MID-TERM PRACTICE PAPER

Session 2026-27 · Term II · Chapters 8–13

Class: 8	Subject: Science	Set 05	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. All questions are compulsory.
2. This question paper has five sections A, B, C, D and E. Marks are indicated against each question.
3. Section A (Q. 1, parts I–XII) consists of multiple choice questions of 1 mark each.
4. Section B (Q. 2–5) consists of very short answer questions of 2 marks each.
5. Section C (Q. 6–8) consists of short answer questions of 3 marks each.
6. Section D (Q. 9–11) consists of long answer questions of 5 marks each.
7. Section E (Q. 12–15) consists of four case-based / source-based questions of 4 marks each (1 + 1 + 2).
8. There is no overall choice. However, internal choice is given in some questions. Only one of the choices in such questions is to be attempted.
9. Draw neat and labelled diagrams wherever required.

SECTION A — Multiple Choice Questions — 1 questions · 12 marks

Question 1 has twelve parts (I–XII). Each carries 1 mark.

1. From Question I–XII, choose the correct option.

I. Which of the following is a mixture? 1

(a) Carbon dioxide (b) Iron
(c) Distilled water (d) Sea water

II. The solvent in a solution of sugar in water is: 1

(a) sugar (b) both (c) none of these (d) water

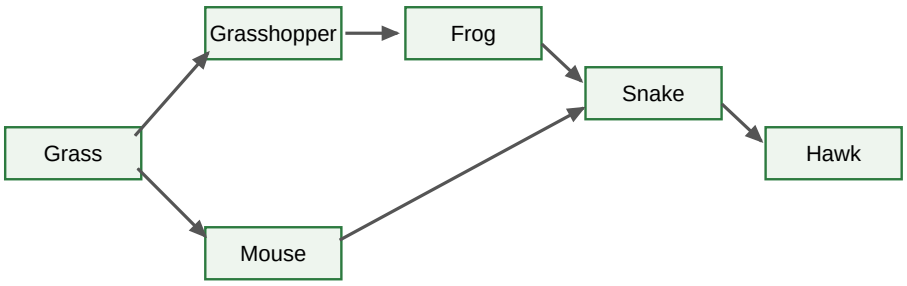
III. The image of an object kept in front of a plane mirror is formed: 1

(a) in front of the mirror at half the distance
(b) at infinity
(c) on the mirror
(d) behind the mirror at the same distance as the object

IV. In a food chain, grass → grasshopper → frog → snake, the secondary consumer is the: 1

(a) snake (b) grasshopper (c) grass (d) frog

V. In the food web shown, the hawk gets energy directly from: 1



(a) grass (b) snake (c) mouse (d) frog

VI. Which organisms break down dead plants and animals and return nutrients to the soil? 1

(a) Herbivores (b) Carnivores (c) Producers (d) Decomposers

VII If all frogs in the food web disappear, which effect is MOST likely? 1

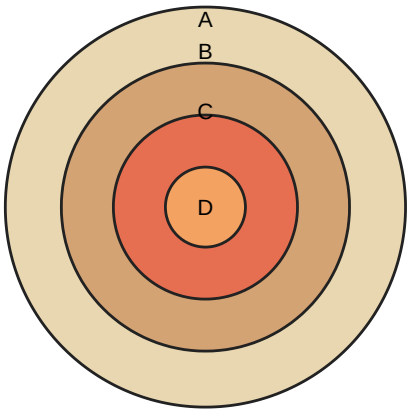
(a) Grass numbers rise, grasshoppers fall

(b) Hawks multiply rapidly

(c) Nothing changes

(d) Grasshopper numbers rise and snake feeds more on mice

VIII In the figure, the layer labelled A is the: 1



(a) crust (b) inner core (c) outer core (d) mantle

IX Which layer of the atmosphere contains the ozone layer that absorbs harmful ultraviolet rays? 1

(a) Mesosphere (b) Stratosphere (c) Troposphere (d) Exosphere

X Seasons on the Earth are caused mainly by: 1

(a) distance of Earth from the Moon

(b) the tilt of the Earth's axis as it revolves around the Sun

(c) phases of the Moon

(d) rotation of the Earth

XI The Moon takes about _____ to complete one phase cycle from one new moon to the next. 1

(a) 365 days (b) 7 days (c) 15 days (d) 29.5 days

XII A simple pendulum completes 20 oscillations in 40 s. Its time period is: 1

(a) 0.5 s (b) 800 s (c) 2 s (d) 20 s

SECTION B — Very Short Answer Questions — 4 questions · 8 marks

2 marks each (about 30–40 words).

2. What is a food chain? Give an example of a food chain with three organisms. **2**

OR

Define producer and consumer with an example each.

3. Name the four spheres of the Earth. **2**

4. What is meant by the time period of a pendulum? Give its SI unit. **2**

5. Why is the Moon visible although it does not produce light? **2**

SECTION C — Short Answer Questions — 3 questions · 9 marks

3 marks each (about 60 words).

6. How do decomposers help in the ecosystem? Give two examples. **3**

OR

Explain why cutting of forests disturbs the balance of an ecosystem. Give two points.

7. Distinguish between rotation and revolution of the Earth and state one result of each. **3**

8. What are the four layers of the Earth? Arrange them from the surface to the centre. **3**

OR

Describe the water cycle in three steps.

SECTION D — Long Answer Questions — 3 questions · 15 marks

5 marks each.

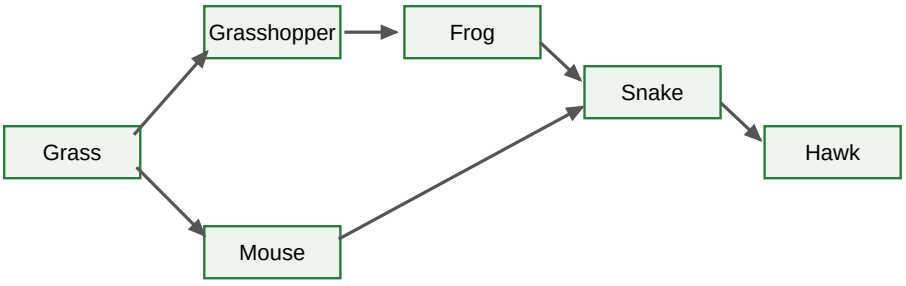
9. Complete the given sentences (choose from the words in the box): **(Ecosystem, Biosphere, Lunar, Producers, Pendulum)** **5**

i. Green plants are _____ because they make their own food.
ii. A community of living organisms together with its non-living surroundings is an _____.
iii. A _____ month is based on phases of the Moon.
iv. The _____ is the zone of the Earth where life exists.
v. A _____ swings to and fro and can be used to measure small intervals of time.

10. Match the Columns **5**

Column I	Column II
(a) Crust	(i) lowest layer of the atmosphere with weather
(b) Troposphere	(ii) outermost layer of the Earth
(c) Producer	(iii) uses sunlight to make food
(d) Full moon	(iv) Moon's whole face is lit as seen from Earth
(e) Pendulum	(v) bob that oscillates to and fro
	(vi) fixed star

11. (a) With a diagram, show the food web of five organisms and name one herbivore. (3) **5**
(b) What will happen to the grasshoppers if the frogs are removed? (2)



OR

(a) Explain how the tilt of the Earth's axis causes unequal day and night (3)
(b) Why is summer hotter than winter in the same place? (2)

SECTION E — Case-based / Source-based Questions — 4 questions · 16 marks

Read the cases carefully and answer. Each question carries 4 marks.

12. A pond has water plants, small fish, large fish, frogs, insects, and bacteria at the bottom. A farmer sprays heavy pesticide near it and many small fish die.

(a) Name the producers in the pond. 1

(b) Which organisms act as decomposers? 1

(c) What will happen to the number of large fish if the small fish decrease? 2

OR

(d) Suggest two ways to protect this ecosystem.

13. Arjun made a pendulum using a string of 1 m and a stone. He counted 10 oscillations in 20 s. Then he shortened the string to 0.5 m.

(a) What is the time period of the first pendulum? 1

(b) What is the frequency of the first pendulum? 1

(c) What will happen to the time period when the string is shortened? 2

OR

(d) Why can pendulum clocks be used to measure time?

14. The diagram of the Moon shows four positions P, Q, R and S in a month. At P no part of the Moon facing Earth is lit and at R the whole face is lit.



P Q R S

(a) Name the phase at P. 1

(b) Name the phase at R. 1

(c) Why do we see different phases of the Moon? 2

OR

(d) Why do we always see the same face of the Moon from the Earth?

15. In a hill area, a farmer cuts trees on the slopes. In the next rainy season, rain washes soil downhill, springs dry up and many birds disappear.

(a) Which sphere of the Earth is directly disturbed by cutting trees? 1

(b) What is soil erosion? 1

(c) Explain how forest loss affects the water cycle. 2

OR

(d) Suggest two actions to restore the hill ecosystem.